

Laroco, Stephen Kyle V.	September 13, 2025
Week 08 - Laboratory - CRISP-DM (Data Understanding and Data Preparation)	Mr. Richard Kho

Showing rows 0 - 24 (45211 total. Query took 0.0004 seconds.)

```
SELECT * FROM `bank_marketing`;
```

[Profiling](#)
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1

> >>

Number of rows: 25

Filter rows:

Sort by key:

None

Extra options

☐				1	58	management	married	tertiary	no	2143.00	yes	no	unknown	5 may	261	1	-1	0	unknown	no
☐				2	44	technician	single	secondary	no	29.00	yes	no	unknown	5 may	151	1	-1	0	unknown	no
☐				3	33	entrepreneur	married	secondary	no	2.00	yes	yes	unknown	5 may	76	1	-1	0	unknown	no
☐				4	47	blue-collar	married	unknown	no	1506.00	yes	no	unknown	5 may	92	1	-1	0	unknown	no
☐				5	33	unknown	single	unknown	no	1.00	no	no	unknown	5 may	198	1	-1	0	unknown	no
☐				6	35	management	married	tertiary	no	231.00	yes	no	unknown	5 may	139	1	-1	0	unknown	no
☐				7	28	management	single	tertiary	no	447.00	yes	yes	unknown	5 may	217	1	-1	0	unknown	no
☐				8	42	entrepreneur	divorced	tertiary	yes	2.00	yes	no	unknown	5 may	380	1	-1	0	unknown	no
☐				9	58	retired	married	primary	no	121.00	yes	no	unknown	5 may	50	1	-1	0	unknown	no
☐				10	43	technician	single	secondary	no	593.00	yes	no	unknown	5 may	55	1	-1	0	unknown	no
☐				11	41	admin.	divorced	secondary	no	270.00	yes	no	unknown	5 may	222	1	-1	0	unknown	no
☐				12	29	admin.	single	secondary	no	390.00	yes	no	unknown	5 may	137	1	-1	0	unknown	no
☐				13	53	technician	married	secondary	no	6.00	yes	no	unknown	5 may	517	1	-1	0	unknown	no
☐				14	58	technician	married	unknown	no	71.00	yes	no	unknown	5 may	71	1	-1	0	unknown	no
☐				15	57	services	married	secondary	no	162.00	yes	no	unknown	5 may	174	1	-1	0	unknown	no
☐				16	51	retired	married	primary	no	229.00	yes	no	unknown	5 may	353	1	-1	0	unknown	no
☐				17	45	admin.	single	unknown	no	13.00	yes	no	unknown	5 may	98	1	-1	0	unknown	no
☐				18	57	blue-collar	married	primary	no	52.00	yes	no	unknown	5 may	38	1	-1	0	unknown	no
☐				19	60	retired	married	primary	no	60.00	yes	no	unknown	5 may	219	1	-1	0	unknown	no
☐				20	33	services	married	secondary	no	0.00	yes	no	unknown	5 may	54	1	-1	0	unknown	no
☐				21	28	blue-collar	married	secondary	no	723.00	yes	yes	unknown	5 may	262	1	-1	0	unknown	no
☐				22	56	management	married	tertiary	no	778.00	yes	no	unknown	5 may	154	1	-1	0	unknown	no

CRISP-DM: STEP 2 | DATA UNDERSTANDING

REFLECTION:

1. What challenges did you experience during the MySQL data-dump restoration?

Answer: I wasn't able to follow what's on the instructions because I faced technical difficulties so I used another way that was taught by dr. larry vea, I used xampp where I edit config so I can import large datasets. From xampp mysql admin, I then went to the import section then imported the file "20250913-Laboratory.sql". I think not being able to follow the instructions is okay as long as you were able to execute the task.

2. With respect to the to the newly created MySQL table, complete the details below:

Number of attributes:	18
Number of objects:	45211

3. With respect to the objects inside the table, complete the details below

Attribute	Data Type
id	int(11)
age	int(11)
job	varchar(50)
marital	varchar(20)
education	varchar(50)
default_status	varchar(10)
balance	decimal(10,2)
housing	varchar(5)
loan	varchar(5)
contact	varchar(20)
day	trying(3)
month	varchar(20)
duration	int(11)
campaign	int(11)
pdays	int(11)
previous	int(11)
outcome	varchr(20)
target	varchar(20)

CRISP-DM: PART 3 | DATA PREPARATION

1. Provide the initial summary for the following attributes:

a. Job:

Job	count
???	11
Adm	175
admin.	4996
blue-collar	9732
entrepreneur	1487
housemaid	1240
management	9458
Ret	94
retired	2170
self-employed	1579
services	3639
student	938
svcs	515
technician	7597
unemployed	1303
unknown	277

b. Marital

Marital	count
Div	583
divorced	4624
married	27214
single	12790

c. Education

Education	count
2nd	120
3rd	29
???	14
primary	6851
secondary	23082
tertiary	13272
unknown	1843

d. Contact

Contact	count
cell.	338
cellular	28947
Tel	187
telephone	2719
unknown	13020

2. Provide the final summary for the following attributes:

a. Job:

Job	count
Adm	175
admin.	4996
blue-collar	9732
entrepreneur	1487
housemaid	1240
management	9458
Ret	94
retired	2170
self-employed	1579
services	3639
student	938
svcs	515
technician	7597
unemployed	1303
unknown	288

b. Marital:

marital	freq ▾ 1
married	27214
single	12790
divorced	4624

c. Education:

education	freq ▾ 1
secondary	22635
tertiary	13186
primary	6821
unknown	1837
2nd	120
3rd	29

d. Contact

contact	freq ▾ 1
cellular	28976
unknown	12758
telephone	2894

3. Based on the results produced in #1 and #2, what issues did you encounter during the clean-up process?

- While doing the cleanup, I noticed a lot of inconsistencies in the dataset. In the job column, there was “???” and I standardized it and joined it to “unknown”. In the marital column, there was “div” which actually meant divorced, so I had to standardize it. For education, there were very fragmented categories like “basic.4y” and “basic.6y” that I needed to combine into just “basic,” and there were also some unclear inputs like “???” that I had to turn into “unknown.” In the contact column, there were abbreviations such as “cell.” and “tel” that I converted into “cellular” and “telephone.” The challenge was mainly deciding how to properly group or rename these values without losing important information, and making sure the categories stayed consistent.

4. What are the possible repercussions if data clean-up is neglected during the “Data Preparation” phase of the Crisp-DM framework?

- If data cleaning is skipped, it can cause a lot of problems in the analysis. For example, “admin” and “admin.” would be treated as different jobs even though they mean the same thing, which would give wrong frequency counts. Having “div” separate from “divorced” could also lead to confusion in marital status analysis. The education column would end up being too fragmented, and this could make it hard to see actual patterns. Inconsistent contact values would make it unreliable to check which communication channel is most common. Overall, if data preparation is neglected, it lowers the quality of the dataset, creates misleading results, and can affect the accuracy of models and decisions based on the data.